

NEAR: NDN Wireless AR Platform

Jinghao Zhao

UCLA



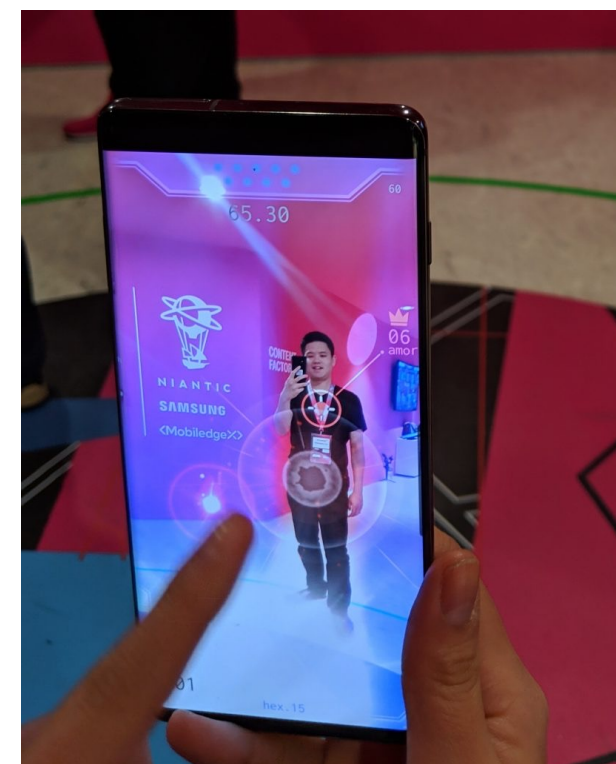
Prevailing AR with Edge

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- Mobile Edge Computing (MEC) provides new opportunities for Augmented Reality (AR). The edge AR system usually has the following requirements:
 - Support various AR applications
 - Handle high dynamic from user mobility, workload, and applications.
 - High requirements for network latency & bandwidth



Remote Training



Mobile AR Games



Industrial AR Wearables

Limitations of Current AR

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- **Multi-layered Indirection:** Hard to access content and application-level information from the network system side, and vice versa.
- **Burdensome Design:** Various security schemes, network stacks, processing, data storages, etc., are deployed for AR modules even in the same edge server, which is hard to manage & prone to error.
- **Do Not Scale:** Unicast-based delivery over broadcast wireless medium. Lack scalability to support massive connections such as commencement & theme park.

Motivations

- We want to build the AR platform that supports:
 - Data centric transmission with built-in multicast support
 - Integrate the NDN wireless with cross-layer components including AR applications, acceleration, security, etc.
 - Extensible for future development
 - Provide an open-source testbed for performance comparison

NEAR Platform

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We present the NEAR (NDN wireless Augmented Reality) Platform, a platform that shows the integrated system, which includes:

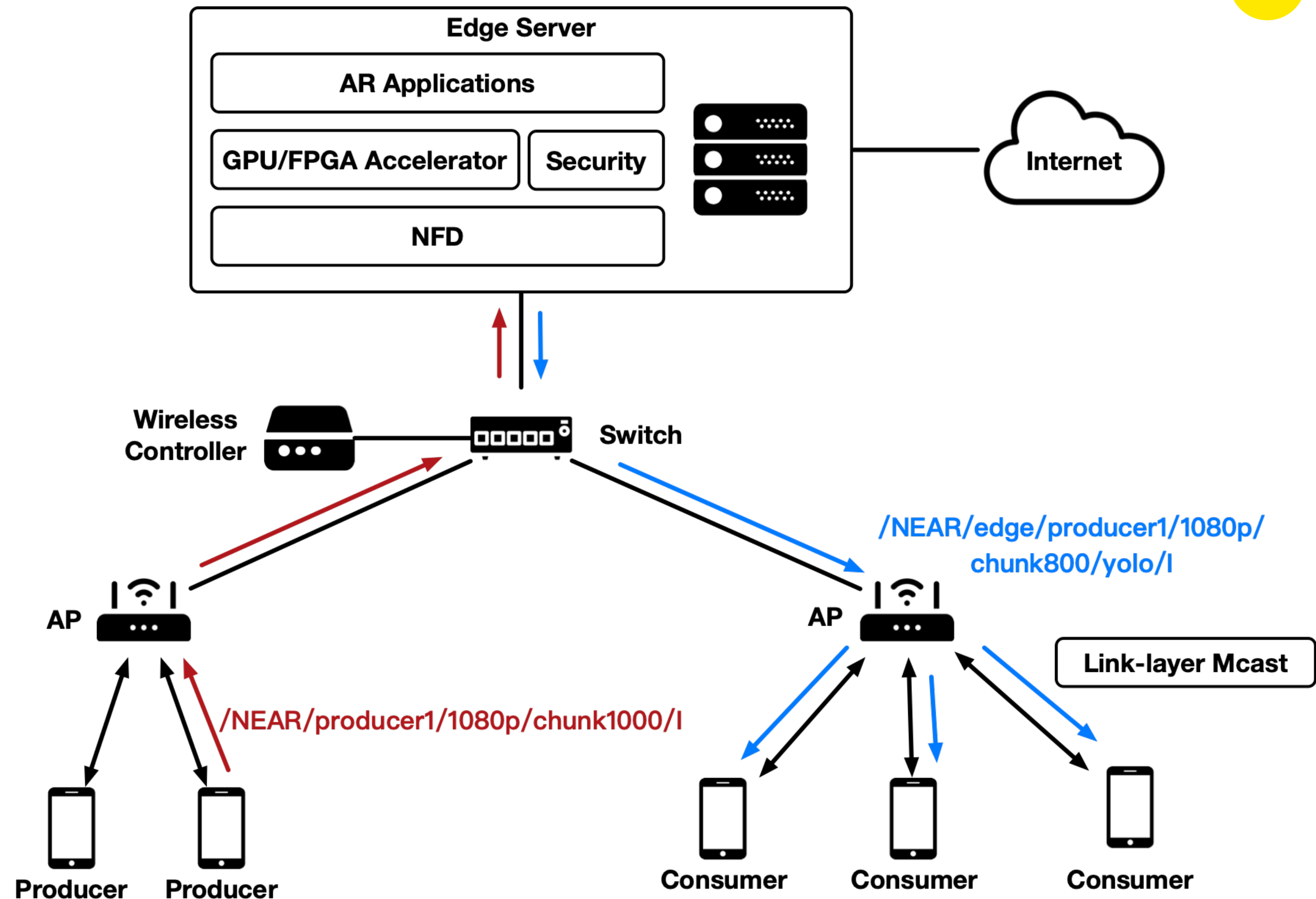
- Application: Heterogeneous AR tasks
- Network: NDN-Wireless with link-layer multicast
- Acceleration: GPU/FPGA-based accelerator for AR tasks
- Security: Naming-based Security Integration

* NEAR Platform will be released on <http://metro.cs.ucla.edu/NEAR.html>

NEAR Overview

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- NEAR platform contains the following components:
 - Producers keep publishing real-time camera views
 - Edge server retrieves camera-views and performs various AR processing with accelerators
 - Consumers retrieve AR contents with link-layer multicast
 - Access points supporting NDN wireless with link-layer mcast
 - Integrated NDN Security

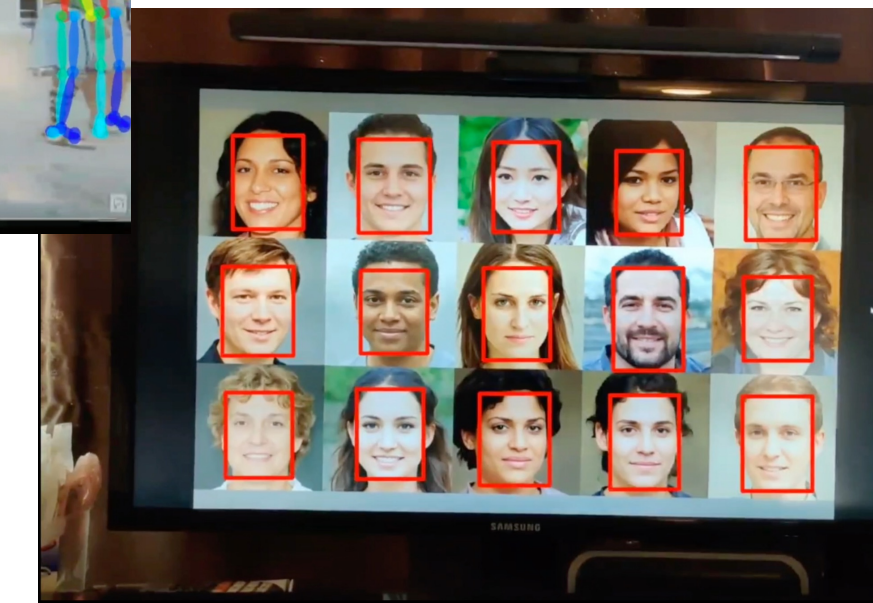
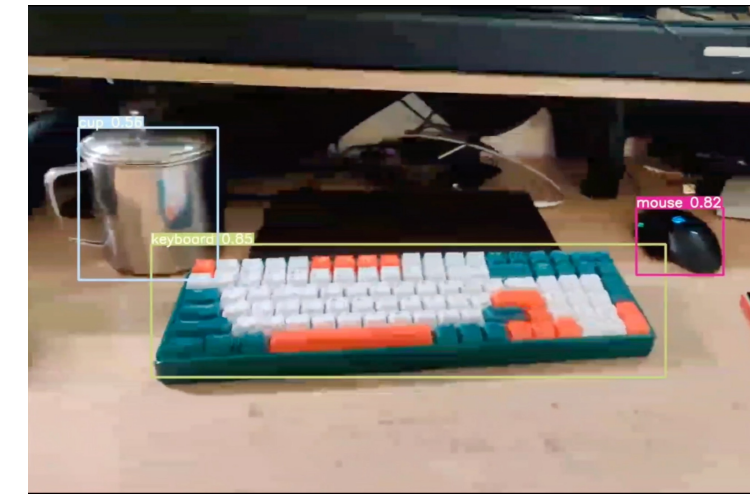


AR Modules

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We have designed and developed a variety of AR modules :

- Object detection & Face recognition & Pose detection
- 3D model rendering
- AR video overlay
- Share panorama view between users
- Chroma key video for interactive opera



AR Requirements

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- To ensure high-quality AR experiences with 2D contents, the bandwidth requirements for each client needs 40Mbps. Considering recent volumetric streaming which streams 3D AR models, each client needs 100Mbps to ensure 30FPS payout.
- AP supports current smartphones with 2x2 MIMO can only support limited clients for the volumetric AR:
 - 802.11n: 300Mbps PHY rate , up to 3 users
 - 802.11ac: 866.7 Mbps PHY rate, up to 8 users
 - 802.11ax: 1201 Mbps PHY rate, up to 12 users
- Due to protocol overhead & noise, devices cannot reach the highest rate all the time. The supported number will be further reduced. **How can we scale the AR support for scenarios involve a large amount of devices?**

NDN Wireless

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- NDN naturally supports multicast & benefits the scalability. However, current link-layer still performs device-centric unicast-based delivery over broadcast wireless medium. Multicast at higher layers will be converted to multiple-unicast at the link layer.
- To fully enable the data centric design at cross layer, NEAR integrates the NDN wireless, which leverages the high-performance link-layer multicast as the basic transmission scheme:
 - Leverage the stateful data plane of NDN
 - Link-layer multicast up to 450Mbps & peak application speed ~100Mbps

Acceleration Integration

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- NEAR supports both GPU & FPGA as the accelerator. Currently the FPGA accelerator could be used for Yolo and OpenPose tasks as a showcase of acceleration-as-a-service in edge computing.
- After receiving the real-time producer frame, the CNN computation part of OpenPose is offloaded to the FPGA/GPU. Once the results are ready, they are used to generate AR contents with skeletons, which is published to all consumers through NDN.



GPU Accelerator



FPGA-based Accelerator

Security Integration

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- NEAR also integrate the security into NEAR following NDN specifications, including:
 - Authentication
 - Integrity protection
 - AR content Encryption
 - Access control with consumer's certificates

How AR work with NEAR?

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- Giving the benefits of multicast, directly applying multicast to AR has the following issues:
 - How to handle the AR frame loss?
 - How to fully leverage multicast & reduce missed interest compression?
 - How to design the security schemes for AR contents?

Issue 1: Frame Loss

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- We first tests previous video streaming namespace that split stream with I/P frames. However, I frames are frequently lost and blocks the AR streaming.
- **Why High Loss Rate for I-frames?** In AR streams, I-frames are much larger than P-frames. Average I-frame length is ~100000 bytes, while P-frame is ~10000 bytes. Each link-layer frame can contain at most 1480 bytes as the payload. Thus, each I-frame requires ~68 frames while P-frame requires only ~7 frames.
- Multicast do not have any retransmission schemes at lower layers. **Thus, a single link-layer frame lost causes I-frame lost.** Although the application performs the retransmission, the I-frame retransmission still suffers from a high loss rate.

```
Append I frame of length 122609 frame number: 1561 1621097832.862848
Append I frame of length 61933 frame number: 1591 1621097833.8685782
Append I frame of length 103242 frame number: 1621 1621097834.8730826
Append I frame of length 73125 frame number: 1651 1621097835.8784173
Append I frame of length 73727 frame number: 1681 1621097836.8840656
Append I frame of length 59985 frame number: 1711 1621097837.8879087
Append I frame of length 77700 frame number: 1741 1621097838.8922062
Append I frame of length 112228 frame number: 1771 1621097839.8973887
Append I frame of length 104134 frame number: 1801 1621097840.9023616
Append I frame of length 103152 frame number: 1831 1621097841.9075658
Append I frame of length 105217 frame number: 1861 1621097842.9134364
Append I frame of length 129086 frame number: 1891 1621097843.985174
Append I frame of length 125792 frame number: 1921 1621097844.957318
Append I frame of length 103786 frame number: 1951 1621097845.9308825
Append I frame of length 74750 frame number: 1981 1621097846.934015
Append I frame of length 69365 frame number: 2011 1621097847.939368
Append I frame of length 74960 frame number: 2041 1621097848.9443557
Append I frame of length 75821 frame number: 2071 1621097849.94862
Append I frame of length 89735 frame number: 2101 1621097850.9544508
Append I frame of length 88603 frame number: 2131 1621097851.993351
Append I frame of length 89010 frame number: 2161 1621097852.997422
```

I-frame

```
Append P frame of length 14663 frame number: 2329
Append P frame of length 17297 frame number: 2330
Append P frame of length 10163 frame number: 2331
Append P frame of length 16312 frame number: 2332
Append P frame of length 10420 frame number: 2333
Append P frame of length 11909 frame number: 2334
Append P frame of length 444 frame number: 2335
Append P frame of length 20389 frame number: 2336
Append P frame of length 11359 frame number: 2337
Append P frame of length 12445 frame number: 2338
Append P frame of length 8232 frame number: 2339
Append P frame of length 9105 frame number: 2340
Append P frame of length 8109 frame number: 2342
Append P frame of length 6726 frame number: 2343
Append P frame of length 8736 frame number: 2344
Append P frame of length 5437 frame number: 2345
Append P frame of length 5891 frame number: 2346
Append P frame of length 224 frame number: 2347
```

P-frame

NEAR Namespace Design

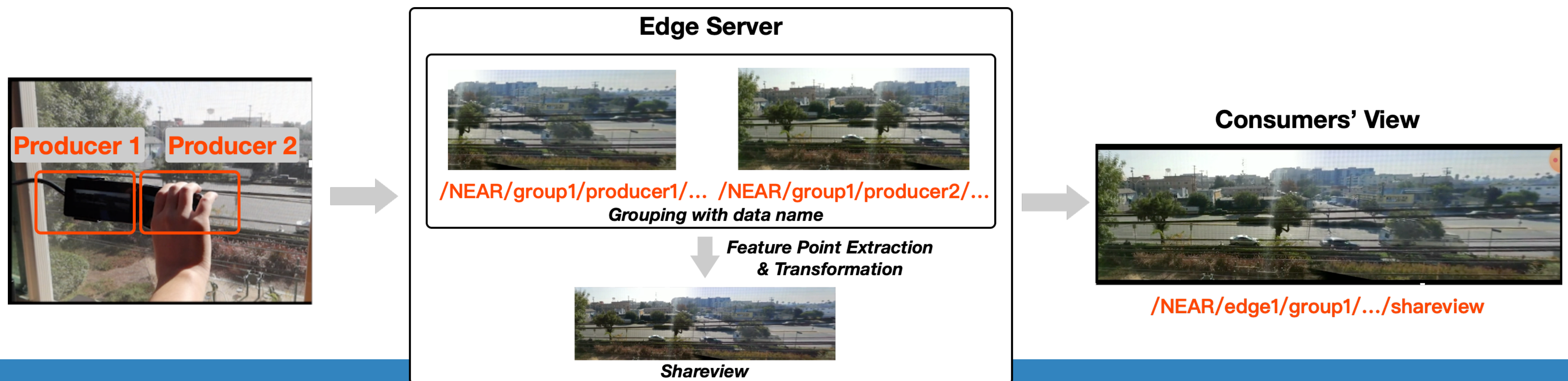
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- **NEAR requires new namespace designs to organize AR data.** To solve issues caused by unbalanced I/P frame sizes, NEAR segments the AR stream with “**chunk**”, which has a uniform content length. NEAR design the namespace for producer streams & AR contents:
 - Producer: /NEAR/{group ID}/{producer ID}/{quality}/{chunk number}/{frame type}
 - E.g., /NEAR/group1/producer1/1080p/1234/I represents chunk 1234 from producer1, which content belongs to the 1080P I frame.
 - AR Contents: /NEAR/{Edge ID}/{group ID}/{producer ID}/{quality}/chunk/{chunk number}/{AR task}/{frame type}
 - E.g., /NEAR/edge1/group1/producer1/1080p/chunk/1000/yolo/I is used to retrieve the producer1's view processed by the Yolo task

Leveraging NEAR name

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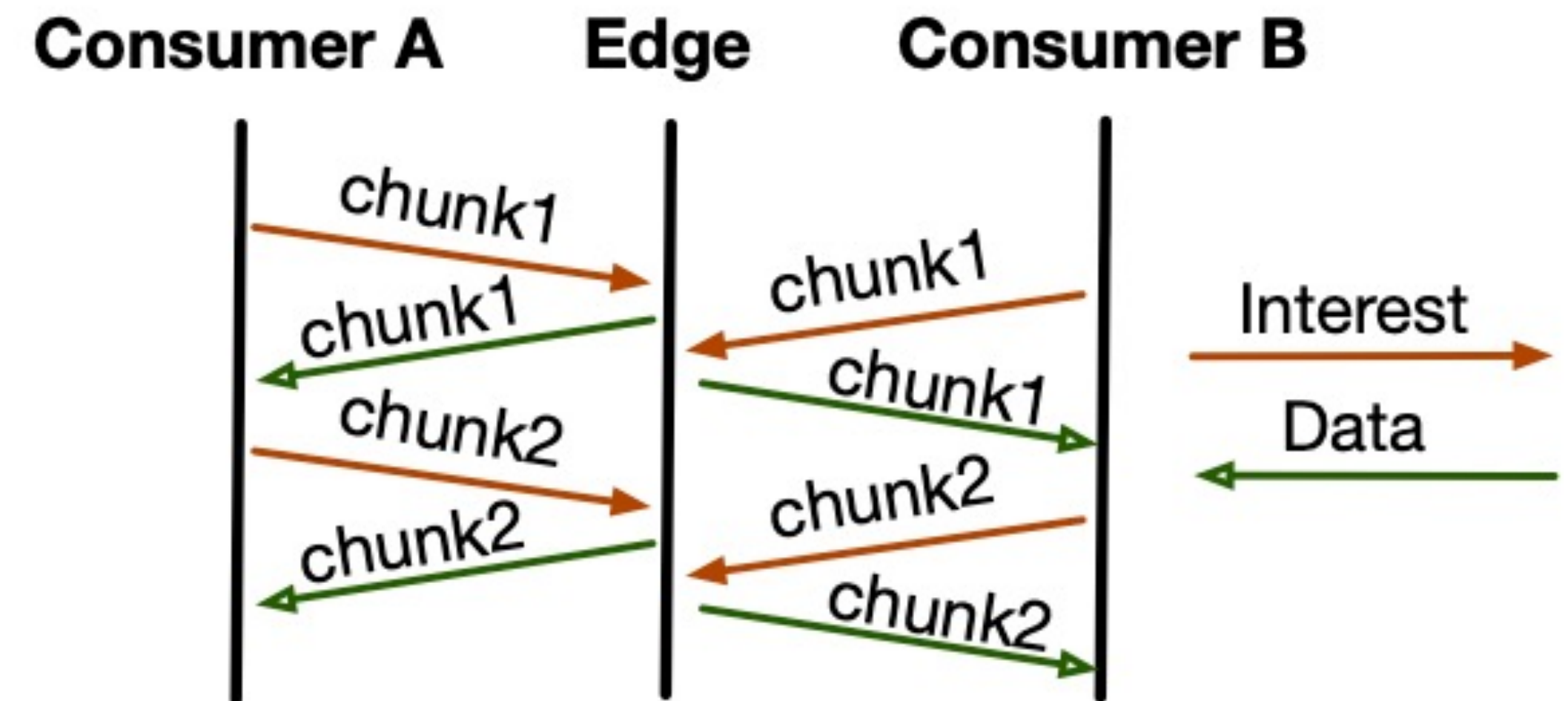
- **NEAR simplify edge AR management:** Consumers retrieve different AR streams with the corresponding AR task in the naming. The edge leverage NDN naming to group consumer interests and perform corresponding processing without extra signaling procedure. Further access control & security is also ensured by NDN without extra application components.
- **NEAR benefits data processing with naming information:** For example, in the shareview task, multiple producers in the same places will share the same group ID in the naming. The data naming enables the edge to cluster producers' views without expensive CV processing.



Issue 2: Frequently Missed Interest Compression

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- When several interests requesting the same content, NFD compresses interests and multicast the data. However, we find **most of the interests are not compressed in the transmission.**
- For example, if the consumer A's interest has been responded before consumer B's interest coming, the same contents will be responded again.

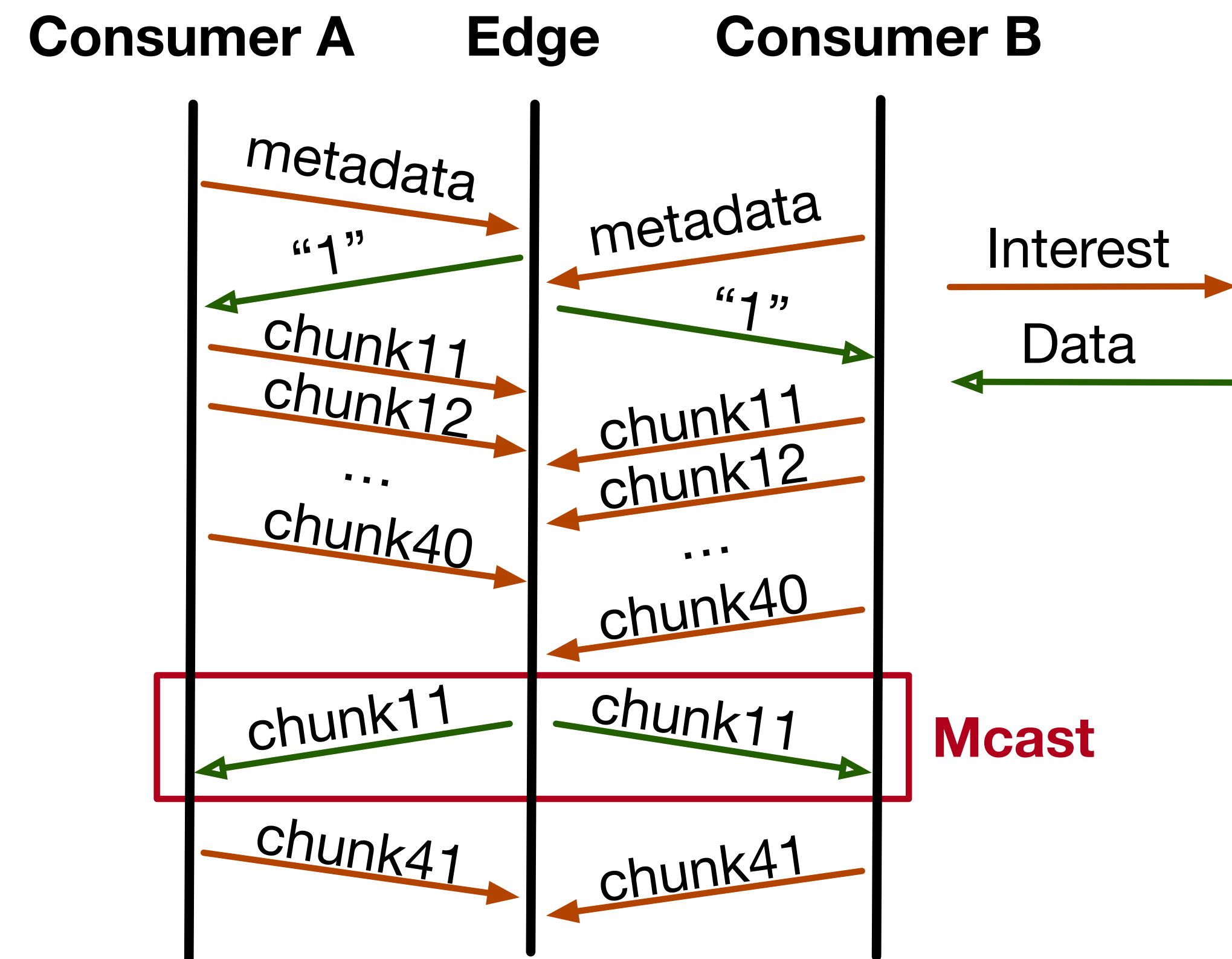


Repeated content transmission when interest compression is missed

NEAR Chunk Prefetching

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- NEAR designs the chunk prefetching scheme to support multiple consumers without sync requirements & ensure interest compression:
 - Each consumer retrieve the AR metadata including latest chunk number at the beginning.
 - After getting metadata, consumers maintain an **outstanding interest window**, which send limited outstanding interests to prefetch future AR contents.
 - Once the AR chunk is published, the data is multicast to all consumers that requested it.
- When the consumer detects packet loss, it resends the interest for retransmission. The requested data is also multicast.



NEAR Design with chunk prefetching (window size 30)

Issue 3: AR contents security

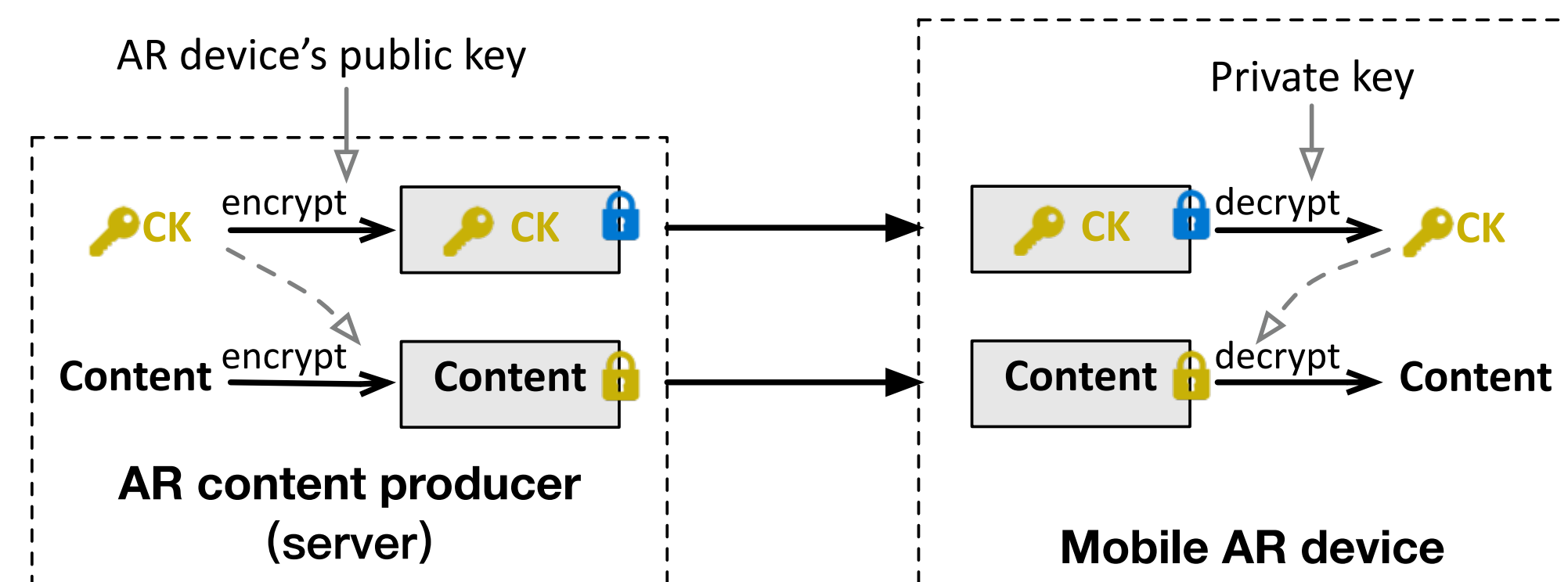
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- We integrate the security into NEAR following NDN specifications. Data packets for both directions (mobile to server, server to mobile) are signed at the producer and verified at the consumer.
 - **Signature:** RSA signature over a SHA-256 digest, following NDN Spec. 0.3, signature type 1.
- It ensures that a packet is created by the claimed producer (authentication/ provenance) and is not tampered with while in transit (integrity). After receiving data, consumer fetches the certificate from the **KeyLocator** if the key is not cached:
 - Name of KeyLocator: “/NEAR/KEY/{Edge ID}/{AR task}” and “/NEAR/KEY/{Producer ID}/{AR task}”
 - Certificate: NDN Certificate Format Version 2.0

Encryption & Access Control

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- **Producers->Edge:** Encrypted by pre-shared content keys with CFB-128 block cipher algorithm
- **Edge->Consumers:** Leverage consumer's public key to exchange content key. Block cipher for subsequent AR content transmission. [Learn from NDN-NAC]
 - Consumer sends interest “/NEAR/server/CK/{AR task}”, to request the content key corresponding AR task. Consumer also includes the its ID & pub-key in the request.
 - Edge checks if consumer's ID & public key in the preset access-control list. If the consumer is allowed to access, the edge responds data with encrypted CK with the consumer's pub-key.
 - Consumer receives CK and decrypts with corresponding private key.
 - All subsequent AR contents are encrypted by CK with CFB-128 block cipher. The IV is updated for every packet to avoid key reuse.



Demo Prototype Setup

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- Edge server
 - Ubuntu 18.04 LTS, Intel i7-9700K CPU, Nvidia RTX 2080S GPU
- Access Point
 - TP-Link N750 router (Wi-Fi chip: Qualcomm Atheros AR9580, OpwnWrt 19.07)
 - TP-Link AC1750 router (Wi-Fi chip: Qualcomm Atheros QCA9558, OpwnWrt 19.07)
- Mobile Clients
 - Google Pixel 4a smartphones (Android 11.0. Qualcomm Snapdragon 730G SoC).
 - Google Pixel smartphones (Android 10.0. Qualcomm Snapdragon 821 SoC).
 - Xiaomi MI6 & MIX2 smartphones (Android 10.0. Qualcomm Snapdragon 835 SoC).

Demo Video

- Demo Video Placeholder

Comparison with multiple consumers

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- To test the scalability of NEAR, we compare the performance between NEAR and unicast-based transmission.
- The edge publish the same 2k source in the testing. We involve 4 consumers with 60Mbps background traffic to show the performance under a heavy congestion.

Thanks!

Download NEAR Platform at <http://metro.cs.ucla.edu/NEAR.html>